数值分析第十七次作业

崔正平 2400010714

1. 定义**离散正弦变换 (DST)** 为

$$F_k = \sum_{j=1}^{N-1} f_j \sin\left(\frac{jk\pi}{N}\right), \quad \forall 1 \leqslant k \leqslant N-1,$$

求证: 其逆变换为

$$f_j = \frac{2}{N}\sum_{k=1}^{N-1} F_k \sin\left(\frac{jk\pi}{N}\right), \quad \forall 1 \leqslant j \leqslant N-1.$$

解答.

将 DST 代入可得

$$\begin{aligned}
&\frac{2}{N}\sum_{k=1}^{N-1} F_k \sin\left(\frac{jk\pi}{N}\right)\\
=&\frac{2}{N}\sum_{k=1}^{N-1}\sum_{\ell=1}^{N-1} f_\ell \sin\left(\frac{\ell k\pi}{N}\right)\sin\left(\frac{jk\pi}{N}\right)\\
=&\frac{2}{N}\sum_{\ell=1}^{N-1} f_\ell \sum_{k=1}^{N-1} \sin\left(\frac{\ell k\pi}{N}\right)\sin\left(\frac{jk\pi}{N}\right)\\
=&\frac{1}{N}\sum_{\ell=1}^{N-1} f_\ell \sum_{k=1}^{N-1} \left(\cos\left(\frac{(\ell-j)k\pi}{N}\right) - \cos\left(\frac{(\ell+j)k\pi}{N}\right)\right).
\end{aligned}$$

再注意到 $\forall t \in \mathbb{Z}$ 满足 $t/2N \notin \mathbb{Z}$, 都有

$$\sum_{k=1}^{N-1} \cos\left(\frac{tk\pi}{N}\right) = \frac{1}{2\sin\left(\frac{t\pi}{2N}\right)}\left(\sin\left(\frac{(N-\frac{1}{2})t\pi}{N}\right) - \sin\left(\frac{\frac{1}{2}t\pi}{N}\right)\right) = \frac{1}{2}((-1)^{t-1}-1),$$

故

(1) 当 $\ell = j$ 时,

$$\sum_{k=1}^{N-1} \left(\cos\left(\frac{(\ell-j)k\pi}{N}\right) - \cos\left(\frac{(\ell+j)k\pi}{N}\right)\right) = N-1-\frac{1}{2}((-1)^{2j-1}-1) = N;$$

(2) 当 $\ell \neq j$ 时,

$$\sum_{k=1}^{N-1}\left(\cos\left(\frac{(\ell-j)k\pi}{N}\right)-\cos\left(\frac{(\ell+j)k\pi}{N}\right)\right)=\frac{1}{2}((-1)^{\ell-j-1}-1)-\frac{1}{2}((-1)^{\ell+j-1}-1)=0;$$

从而 $\forall 1 \leqslant j \leqslant N-1$, 都有

$$\frac{2}{N}\sum_{k=1}^{N-1}F_k\sin\left(\frac{jk\pi}{N}\right)=\frac{1}{N}\sum_{\ell=1}^{N-1}f_\ell\sum_{k=1}^{N-1}\left(\cos\left(\frac{(\ell-j)k\pi}{N}\right)-\cos\left(\frac{(\ell+j)k\pi}{N}\right)\right)=\frac{1}{N}f_j\cdot N=f_j.$$

□

2. 定义**离散余弦变换 (DCT)** 为

$$F_k=\frac{1}{2}(f_0+(-1)^k f_N)+\sum_{j=1}^{N-1}f_j\cos\left(\frac{jk\pi}{N}\right),\quad \forall 0\leqslant k\leqslant N,$$

求证: 其逆变换为

$$f_j=\frac{2}{N}\left(\frac{1}{2}(F_0+(-1)^jF_N)+\sum_{k=1}^{N-1}F_k\cos\left(\frac{jk\pi}{N}\right)\right),\quad \forall 0\leqslant j\leqslant N.$$

解答.

将 DCT 代入可得

$$\begin{aligned}
&\frac{2}{N}\left(\frac{1}{2}(F_0+(-1)^jF_N)+\sum_{k=1}^{N-1}F_k\cos\left(\frac{jk\pi}{N}\right)\right)\\
=&\frac{2}{N}\left\{\frac{1}{2}\left[\frac{1}{2}(f_0+f_N)+\sum_{\ell=1}^{N-1}f_\ell+(-1)^j\left(\frac{1}{2}(f_0+(-1)^Nf_N)+\sum_{\ell=1}^{N-1}f_\ell(-1)^\ell\right)\right]\right.\\
&\left.+\sum_{k=1}^{N-1}\left[\frac{1}{2}(f_0+(-1)^kf_N)+\sum_{\ell=1}^{N-1}f_\ell\cos\left(\frac{\ell k\pi}{N}\right)\right]\cos\left(\frac{jk\pi}{N}\right)\right\}\\
=&\frac{2}{N}\left\{\sum_{\ell=1}^{N-1}f_\ell\left[\sum_{k=1}^{N-1}\cos\left(\frac{\ell k\pi}{N}\right)\cos\left(\frac{jk\pi}{N}\right)+\frac{1}{2}+\frac{1}{2}(-1)^{j+\ell}\right]\right.\\
&+f_0\left[\frac{1}{2}\sum_{k=1}^{N-1}\cos\left(\frac{jk\pi}{N}\right)+\frac{1}{2}\left(\frac{1}{2}+\frac{1}{2}(-1)^j\right)\right]\\
&\left.+f_N\left[\frac{1}{2}\sum_{k=1}^{N-1}(-1)^k\cos\left(\frac{jk\pi}{N}\right)+\frac{1}{2}\left(\frac{1}{2}+\frac{1}{2}(-1)^{N+j}\right)\right]\right\}\\
=&\frac{2}{N}\left(\sum_{\ell=1}^{N-1}f_\ell S_{\ell j}+\frac{1}{2}f_0S_{0j}+\frac{1}{2}f_NS_{Nj}\right),
\end{aligned}$$

其中 $\forall 0 \leqslant \ell, j \leqslant N$,

$$\begin{aligned} S_{\ell j} &= \sum_{k=1}^{N-1} \cos\left(\frac{\ell k\pi}{N}\right)\cos\left(\frac{jk\pi}{N}\right) + \frac{1}{2} + \frac{1}{2}(-1)^{j+\ell} \\ &= \frac{1}{2}\left[\sum_{k=1}^{N-1}\left(\cos\left(\frac{(\ell+j)k\pi}{N}\right) + \cos\left(\frac{(\ell-j)k\pi}{N}\right)\right) + 1 + (-1)^{j+\ell}\right]. \end{aligned}$$

类似第 1 题分类讨论可得

$$S_{\ell j} = \begin{cases} 0, & \ell \neq j; \\ \dfrac{N}{2}, & \ell = j \neq 0, N; \\ N, & \ell = j = 0, N; \end{cases}$$

故 $\forall 0 \leqslant j \leqslant N$, 都有

$$\frac{2}{N}\left(\frac{1}{2}(F_0 + (-1)^j F_N) + \sum_{k=1}^{N-1} F_k \cos\left(\frac{jk\pi}{N}\right)\right) = \frac{2}{N}\left(\sum_{\ell=1}^{N-1} f_\ell S_{\ell j} + \frac{1}{2} f_0 S_{0j} + \frac{1}{2} f_N S_{Nj}\right) = f_j.$$

□

3. 定义**离散半波余弦变换 (Discrete Quarter-Wave Cosine Transform)** 为

$$F_k = \sum_{j=0}^{N-1} f_j \cos\left(\frac{(j+\frac{1}{2})k\pi}{N}\right), \quad \forall 0 \leqslant k \leqslant N-1,$$

求证: 其逆变换为

$$f_j = \frac{2}{N}\left(\frac{1}{2}F_0 + \sum_{k=1}^{N-1} F_k \cos\left(\frac{(j+\frac{1}{2})k\pi}{N}\right)\right), \quad \forall 0 \leqslant j \leqslant N-1.$$

解答.

将离散半波余弦变换代入可得

$$\begin{aligned} &\frac{2}{N}\left(\frac{1}{2}F_0 + \sum_{k=1}^{N-1} F_k \cos\left(\frac{(j+\frac{1}{2})k\pi}{N}\right)\right) \\ =& \frac{2}{N}\left(\frac{1}{2}\sum_{\ell=0}^{N-1} f_\ell + \sum_{k=1}^{N-1}\sum_{\ell=1}^{N-1} f_\ell \cos\left(\frac{(\ell+\frac{1}{2})k\pi}{N}\right)\cos\left(\frac{(j+\frac{1}{2})k\pi}{N}\right)\right) \\ =& \frac{2}{N}\sum_{\ell=0}^{N-1} f_\ell\left(\frac{1}{2} + \sum_{k=1}^{N-1}\cos\left(\frac{(\ell+\frac{1}{2})k\pi}{N}\right)\cos\left(\frac{(j+\frac{1}{2})k\pi}{N}\right)\right) \\ =& \frac{1}{N}\sum_{\ell=0}^{N-1} f_\ell\left[1 + \sum_{k=1}^{N-1}\left(\cos\left(\frac{(\ell-j)k\pi}{N}\right) + \cos\left(\frac{(\ell+j+1)k\pi}{N}\right)\right)\right]. \end{aligned}$$

类似第 1 题分类讨论可得

$$\sum_{k=1}^{N-1}\left(\cos\left(\frac{(\ell-j)k\pi}{N}\right)+\cos\left(\frac{(\ell+j+1)k\pi}{N}\right)\right)=\begin{cases}-1, & \ell\neq j;\\ N-1, & \ell=j;\end{cases}$$

故 $\forall 0\leqslant j\leqslant N-1$, 都有

$$\begin{aligned}&\frac{2}{N}\left(\frac{1}{2}F_0+\sum_{k=1}^{N-1}F_k\cos\left(\frac{(j+\frac{1}{2})k\pi}{N}\right)\right)\\ =&\frac{1}{N}\sum_{\ell=0}^{N-1}f_\ell\left[1+\sum_{k=1}^{N-1}\left(\cos\left(\frac{(\ell-j)k\pi}{N}\right)+\cos\left(\frac{(\ell+j+1)k\pi}{N}\right)\right)\right]\\ =&\frac{1}{N}f_j(1+(N-1))=f_j.\end{aligned}$$

□

4. 通过离散半波余弦变换推导矩阵

$$\boldsymbol{A}=\begin{pmatrix}-1 & 1 & & & \\ 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1\\ & & & 1 & -1\end{pmatrix}_{N\times N}$$

的特征分解形式.

解答.

设 $\boldsymbol{x}=(x_0,x_1,\cdots,x_{N-1})^{\mathrm{T}}\in\mathbb{R}^N\backslash\{\boldsymbol{0}\}$ 满足 $\boldsymbol{Ax}=\lambda\boldsymbol{x}$, 则

$$\begin{cases}x_{j-1}-2x_j+x_{j+1}=\lambda x_j, & \forall 1\leqslant j\leqslant N-2;\\ -x_0+x_1=\lambda x_0;\\ x_{N-2}-x_{N-1}=\lambda x_{N-1}.\end{cases}$$

再引入虚节点 $x_{-1}=x_0,x_N=x_{N-1}$, 则

$$\begin{cases}x_{j-1}-2x_j+x_{j+1}=\lambda x_j, & \forall 0\leqslant j\leqslant N-1;\\ x_{-1}=x_0;\\ x_N=x_{N-1}.\end{cases}$$

$\forall 0\leqslant k\leqslant N-1$, 取离散半波余弦变换的基向量 $\boldsymbol{x}^{(k)}\in\mathbb{R}^N$, 同时也引入虚节点, 即

$$x_j^{(k)}=\cos\left(\frac{(j+\frac{1}{2})k\pi}{N}\right),\quad \forall -1\leqslant j\leqslant N,$$

直接代入上述方程组计算可得

$$\lambda_k = 2\cos\left(\frac{k\pi}{N}\right) - 2 = -4\sin^2\left(\frac{k\pi}{2N}\right),$$

由此可得矩阵 $\boldsymbol{A}$ 的所有特征值和对应的特征向量. $\square$